10. THE HEART : DEVELOPMENT OF ARTERIAL ANATOMY

DURATION : 04:14 STUDY SHEET

COURSE SUMMARY

This video delves into the heart of the embryonic development of blood cells, describing their origin from the umbilical vesicle to the bone marrow. It also explores the persistence of the mesenteric artery, a vital remnant of this process, and its role in intestinal development. The aortic arch is highlighted to illustrate the rhythmic movement of the heart and its impact on blood circulation. By associating cardiac dynamics with the fascial environment that maintains vital structures such as the oesophagus and bronchi, the video reveals how imbalances in this rhythmic zone can influence cardiological and musculoskeletal pathologies. An essential lesson for any therapist interested in bodily interconnections.

THE HEART: DEVELOPMENT OF ARTERIAL ANATOMY

ORIGIN AND DEVELOPMENT OF BLOOD CELLS

The first **blood cells** appear around the **18th day** of embryonic development. Their origin occurs in three important stages.

Initially, they come from the **umbilical vesicle**, where **primitive megaloblasts** are found. This phase lasts for the first two months.

From the beginning of the **second month**, a second movement of blood production emerges from the **liver**.

It is only around the **third month** that some of this production begins to occur in the **bone marrow**.

We can observe here a development from the periphery to the centre: first the **umbilico-pedicular** period, then the **hepatic** period, and finally the bone marrow for **red blood cells**.

THE VESTIGE OF VITELLINE DEVELOPMENT: THE MESENTERIC ARTERY

The only vestige that will remain of this vitelline development is the **mesenteric artery**.

It served as an axis of rotation, a **fulcrum**, for the rotation and development of the **intestinal loop**.

When the **amniotic cavity** forms all around it, the vitelline vesicle, which contains the original cells for circulation, reintegrates.

In part, with the vitelline vesicle, the **embryonic pedicle** will form the **umbilical colon** with the placenta. The vestige that will connect is the **mesenteric system** (the superior vesicle). This axis of rotation is crucial for embryonic development.

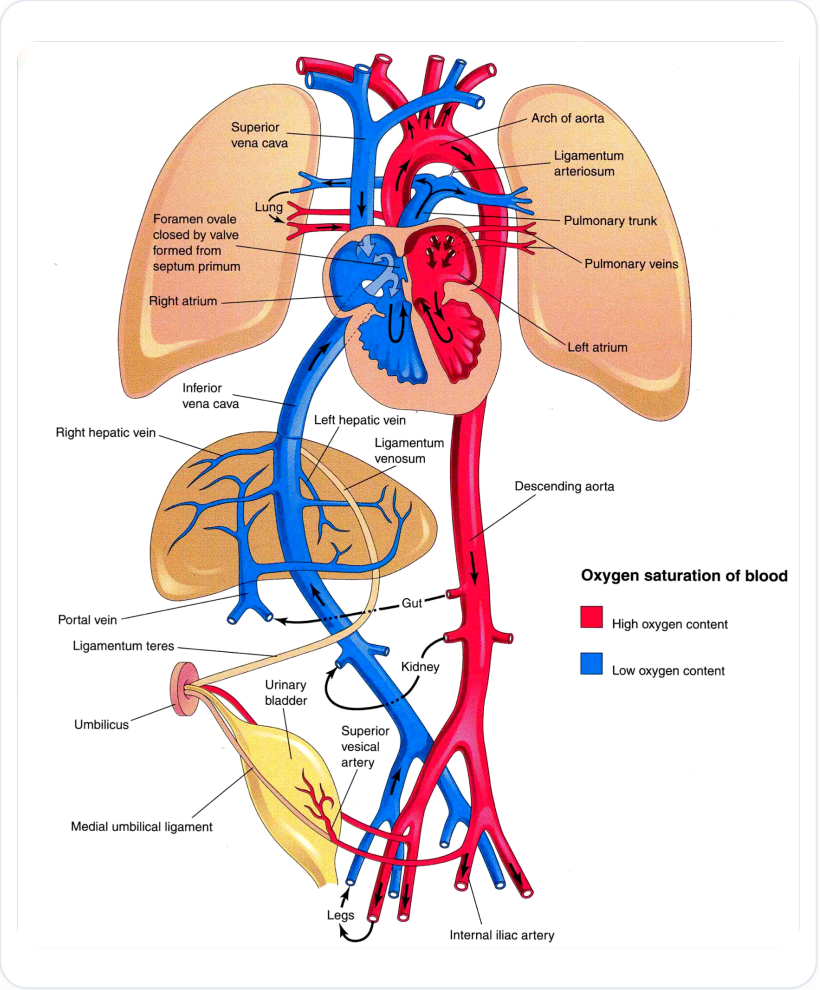


FIG. — Human fetal circulation

THE MOVEMENT OF THE HEART AND THE AORTIC ARCH

When we observe the heart, we see the **aortic arch**. Symbolically, we can say that it ascends or descends.

This movement of the aortic arch represents the **heartbeat**. It almost reproduces a developmental movement.

The aortic arch performs a **twisting** or **untwisting** movement during the cardiac cycle.

When the heart pushes, the aortic arch opens to absorb blood. When the heart empties, it closes, performing a particular movement.

It opens to take blood and closes to eject it, which intrinsically links the aortic arch to the function of the heart.

THE AORTIC ARCH AND THE CARDIAC CYCLE

It's a bit like a **basketball** player. When the heart rises, that's when the aorta ejects its blood volume.

It should be known that the aorta contains almost as much blood in its volume as the heart on the arterial level.

This means that the aorta actively participates in the **systolic ejection** of blood.

This is supported by the **fascial environment**, which could be compared to a **"fishnet stocking"**.

THE RHYTHMIC ZONE: A VITAL FASCIAL CROSSROADS

The aortic arch, the **bronchial bifurcation**, and the **oesophagus** are connected by a common **fascia**, which holds them together.

This area, characterised by the bronchial bifurcation, is what is called the **rhythmic zone point**.

Consider the vital rhythms: **100,000 heartbeats** in 24 hours, **23,000 breaths** in 24 hours, and **1,000 to 2,000 swallows**.

It is a large rhythmic zone that meets and attaches between **T3**, **T4**, and **T5**. These vertebrae form a fascial balance plane for the rhythmic zone, which is both **cardiac**, **pleural**, and **digestive**.

We have the aorta, the heart, and below, a very important bifurcation: the **iliac arteries**. Above, there is yet another level.

In fact, all of this forms a unitary and pure plane. We observe that certain pathologies of the **coxo-femoral** type or others can destabilise the heart. Conversely, what happens in these phases of rotation can manifest up to the level of the heart.

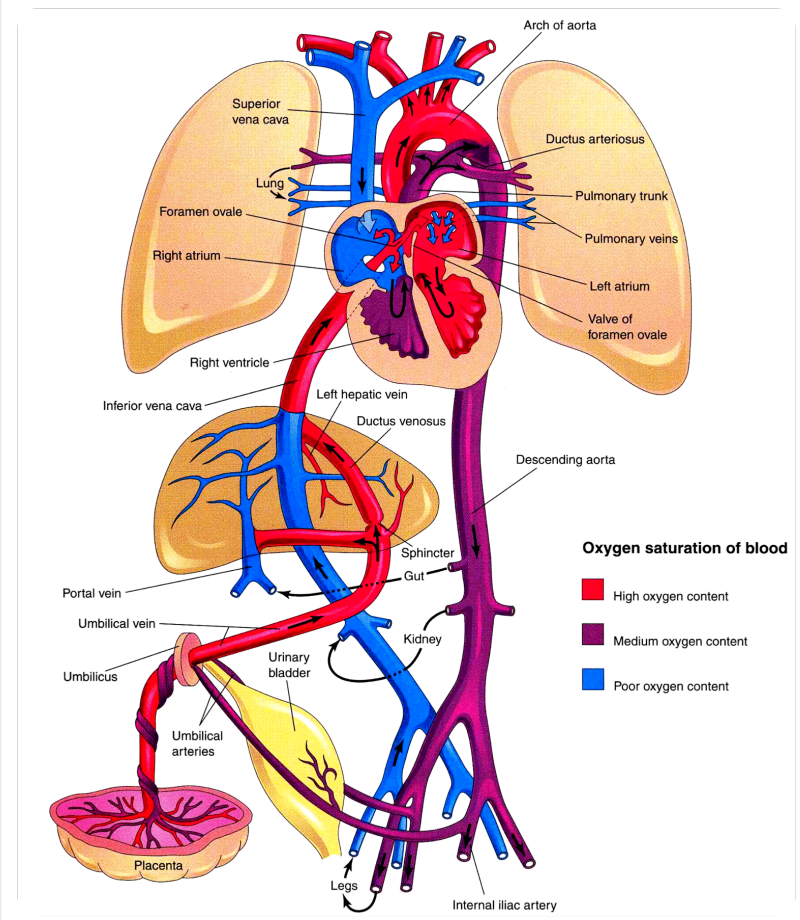


FIG. — Fetal circulation